PRACTICE SHEET

1. The angles of elevation of the top of a tower form the top and foot of a pole are respectively 30^0 and 45^0 is h_T is the height of the tower and h_P is the height of the pole, then which of the following are correct?

1. $\frac{2h_Ph_T}{3+\sqrt{3}} = h_P^2$

2. $\frac{h_T - h_P}{\sqrt{3}+1} = \frac{h_P}{2}$

3. $\frac{2(h_P + h_T)}{h_P} = 4+\sqrt{3}$

Select the correct answer using the code given below:

(a) 1 and 3 only (b) 2 and 3 only
(c) 1 and 2 only (d) 1, 2 and 3

2. From the top of a lighthouse, 100 m high, the angle of depression of a boat is $\tan^{-1}\left(\frac{5}{12}\right)$. What is the distance between the boat and the lighthouse?

(a) 120 m (b) 180 m

(a) 36.3 m (b) 37.3 m
(c) 38.3 m (d) 39.3 m

6. A vertical tower standing on a leveled field is mounted with a vertical flag staff of length 3 m. From a point on the field, the angles of elevation of the bottom and tip of the flag staff are 30^0 and 45^0, respectively. Which one of the following gives the best approximation to the height of the tower?

(a) 3.90 m (b) 4.00 m
(c) 4.10 m (d) 4.25 m

7. The angle of elevation of the top of a tower from a point 20 m away from its base is 45^0. What is the height of the tower?

(a) 10 m (b) 20 m
(c) 30 m (d) 40 m

8. The angles of elevation of the top of a tower standing on a horizontal plane from two points on a line passing through the foot of the tower at distances 49 m and 36 m are 43^0 and 47^0 respectively. What is the height of the tower?